

Study Topics for Building This Portfolio Project

If you want to build a project like this yourself, study these topics in this order.

1. HTML, CSS, and JavaScript basics

- Layout, forms, responsive design, DOM basics, events, and async requests.

2. React

- Components
- Props
- useState
- useEffect
- Form handling
- Conditional rendering
- Rendering lists with map

3. TypeScript basics

- Types and interfaces
- Typing component props
- Typing API responses
- Typing form state

4. Vite and frontend project structure

- Dev server
- Build process
- Environment variables
- Basic project organization

5. REST APIs

- GET vs POST
- JSON request and response flow
- Status codes
- Endpoint design basics

6. Java core

- OOP
- Classes and objects
- Inheritance and interfaces
- Collections
- Exception handling

- Streams
- Generics

7. Spring Boot

- Project setup
- Dependency injection
- @RestController
- @Service
- @Repository
- @Configuration
- Request mapping
- Validation
- application.properties

8. Spring Data JPA and Hibernate

- Entities
- Repositories
- CRUD operations
- JpaRepository
- Table mapping
- In-memory database usage

9. SQL and databases

- Tables
- Primary keys
- Insert, select, update
- Relationships basics

10. Form handling and validation

- Client-side validation
- Backend validation
- Success and error handling

11. Axios or fetch

- Calling APIs from the frontend
- Handling loading, success, and failure states

12. CORS and frontend-backend integration

- Cross-origin requests
- API base URLs
- Local vs production configuration

13. Deployment basics

- Frontend deployment
- Backend deployment
- Environment variables
- Domain setup

Recommended build path:

1. Build a static portfolio first.
2. Convert it to React.
3. Create a Spring Boot backend with one GET endpoint.
4. Connect React to the backend.
5. Add database storage.
6. Add contact form submission.
7. Deploy both parts.

Highest-priority stack for this type of project:

- HTML/CSS/JavaScript
- React
- TypeScript basics
- Java
- Spring Boot
- JPA/Hibernate
- SQL
- REST API integration